

Business Recommendations

Financial Loan Default Prediction Project

1. Use Loan Grade as the Main Risk Check

Grade is the strongest sign of risk in the data. Grade A loans default only 5% of the time, while Grade G loans default almost 58% of the time. The bank should continue using grade as a primary filter when deciding who gets a loan.

2. Watch Borrowers With High Existing Debt

Borrowers who already owe a lot of money compared to their income (high DTI) default much more often — 30% of the time, compared to just 14% for borrowers with low DTI. The bank should add extra checks for these borrowers before approving a new loan.

3. Be More Careful With Certain Loan Purposes

Small business loans default the most (29%). Car and credit card loans default the least (15-16%). The bank could charge higher interest, or ask for more proof of income, for riskier purposes like small business loans.

4. Use the Model to Flag Risky Applications for Manual Review

The prediction model correctly catches about 65 out of every 100 real defaulters. This means the model can act as an early warning system — flagging risky applications so a human reviewer can take a closer look before final approval.

5. Accept Some False Alarms as the Cost of Catching More Risk

The model sometimes flags safe people as risky too (about 34% accuracy on its risk alerts). This is a normal tradeoff — for a bank, missing a real defaulter typically costs more than double-checking a safe applicant.